

3L Construct a String Spelled by a Gapped Genome Path

Gapped Genome Path String Problem

Reconstruct a string from a sequence of (k, d) -mers corresponding to a path in a paired de Bruijn graph.

Input: A collection of (k, d) -mers $(a_1 \mid b_1), \dots, (a_n \mid b_n)$ such that $\text{SUFFIX}(a_i \mid b_i) = \text{PREFIX}(a_{i+1} \mid b_{i+1})$ for all i from 1 to $n - 1$, *PairedReads*.

Output: A string *Text* where the i -th k -mer in *Text* is equal to $\text{SUFFIX}(a_i \mid b_i)$ for all i from 1 to n , if such a string exists.

```
TAA-GCC
AAT-CCA
ATG-CAT
XXXTGC-ATG
XXXXGCC-TGG
      CCA-GGG
XXXXXXCAT-GGA
XXXXXXXXATG-GAT
XXXXXXXXXXTGG-ATG
XXXXXXXXXXXGGG-TGT
XXXXXXXXXXXXGGA-GTT
TAATGCCATGGGATGTT
```

Formatting

Input: Space-separated integers k and d followed by a space-separated list of paired k -mer strings *PairedReads* where individual k -mers within the pair are separated by a "|" character.

Output: A string *Text* where the i -th k -mer in *Text* is equal to $\text{SUFFIX}(a_i \mid b_i)$ for all i from 1 to n , if such a string exists.

Constraints

- The value of k will be between 1 and 10^2 .
- The value of d will be between 1 and 10^3 .
- The number of strings in *PairedReads* will be between 1 and 10^4 .
- The length of any one pair of paired k -mers in *PairedReads* will be between 1 and 10^2 .
- All k -mer strings in *PairedReads* will be DNA strings.

Test Cases

Case 1

Description: The sample dataset is not actually run on your code.

Input:

4 2
GACA|TCTC ACAC|CTCT CACA|TCTC ACAT|CTCA

Output:

GACACATCTCTCA

Case 2

Description: The sample dataset is not actually run on your code.

Input:

2 1
AC|TT CG|TG GT|GA TT|AC

Output:

ACGTTGAC

Case 3

Description: The sample dataset is not actually run on your code.

Input:

2 4
GC|CG CA|GT AT|TG TA|GC AC|CA CC|AT

Output:

GCATACCGTGAT

Case 4

Description: The sample dataset is not actually run on your code.

Input:

5 1
ACAGC|GCGAA CAGCT|CGAAT AGCTG|GAATC GCTGC|AATCA

Output:

ACAGCTGCGAATCA

Case 5

Description: A larger dataset of the same size as that provided by the randomized autograder.